

Model Comparison & Explainable Prediction

Predicting Titanic passenger survival — Logistic Regression vs Random Forest (Python, scikit-learn)

Notebook link: https://colab.research.google.com/drive/1mZpNCqd74_9ZjzdLLsQvTDqh-PUqMcci

1. Dataset overview

The Titanic dataset (loaded with `seaborn.load_dataset("titanic")`) has 891 passengers and 15 columns. The task is binary classification: did the passenger survive (1) or not (0)? It is a common beginner dataset because it is small, easy to understand, and has both real missing values and categorical columns to handle.

Item	Detail
Rows / columns	891 / 15
Target	survived — 61.6% died, 38.4% survived (imbalanced)
Features used	pclass, sex, age, sibsp, parch, fare, embarked
Missing values	age 177 (19.9%), embarked 2, deck 688 (77%, dropped)
Dropped columns	deck (mostly empty); alive, class, who, adult_male, embark_town, alone (duplicate or leak the target)

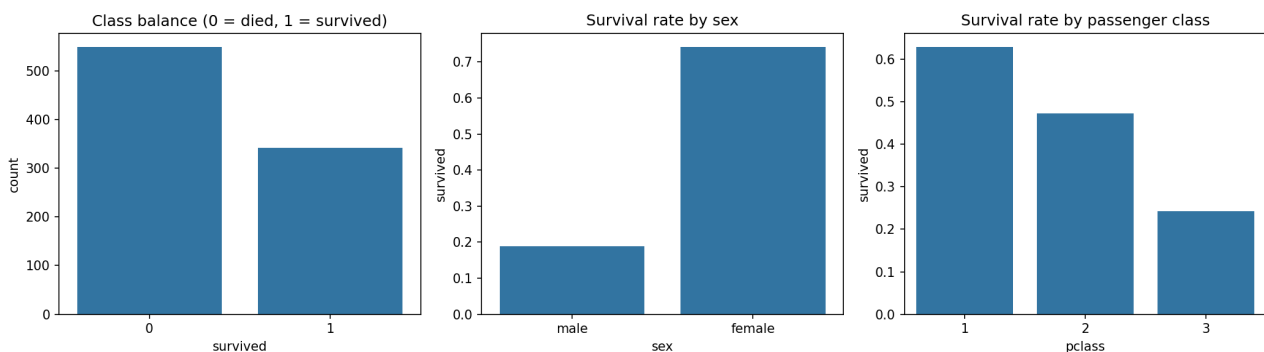


Figure 1. Class balance and survival rate by sex and passenger class. Sex and class are clearly the strongest signals.

2. Workflow

Preprocessing, training and evaluation are kept in separate notebook sections. All preprocessing sits inside a scikit-learn Pipeline, so imputers and scalers learn only from the training data (no test-set leakage).

- **Split:** 80% train (712) / 20% test (179), stratified on the target, `random_state=42` for reproducibility.
- **Missing values:** age → median; embarked → most frequent.
- **Encoding / scaling:** one-hot encoding for sex and embarked; `StandardScaler` on numeric columns.
- **Models:** Logistic Regression (`max_iter=1000`) and Random Forest (200 trees, `max_depth=6`).
- **Evaluation:** accuracy, precision, recall, F1 on the held-out test set, confusion matrices, plus 5-fold cross-validated F1 on the training set as a stability check.

3. Model comparison — metrics

Model	Accuracy	Precision	Recall	F1-score	CV F1 (mean ± std)
Logistic Regression	0.804	0.793	0.667	0.724	0.726 ± 0.034
Random Forest	0.816	0.875	0.609	0.718	0.738 ± 0.028

Precision, recall and F1 are for the positive class (Survived). Selected model highlighted.

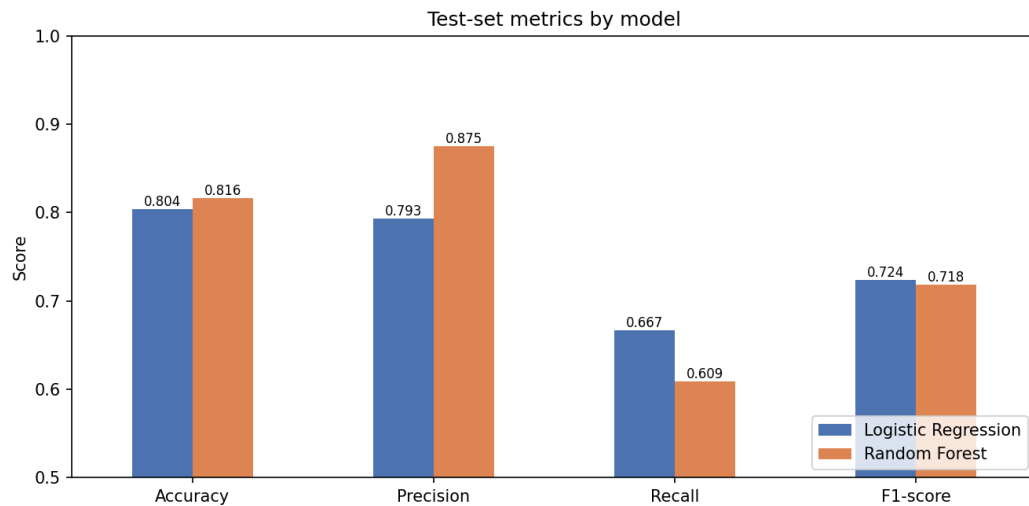


Figure 2. Test-set metrics side by side.

Confusion matrices

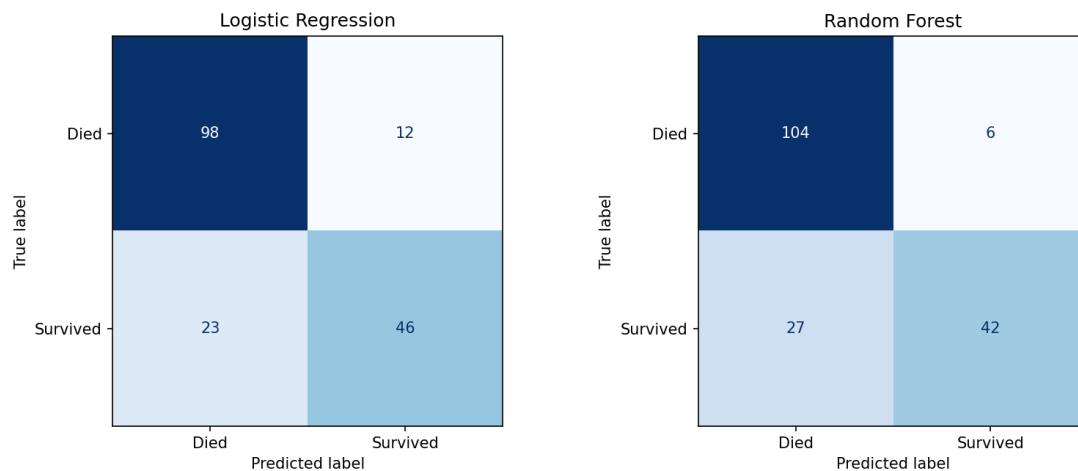


Figure 3. Confusion matrices on the 179 test passengers.

Reading the matrices: Random Forest is more cautious — it predicts "Survived" less often, so it makes only 6 false alarms (vs 12) but misses 27 real survivors (vs 23). Logistic Regression catches 46 of 69 survivors; Random Forest catches 42. That is the precision/recall trade-off in the table.

4. Why I Selected This Model

Selected: Logistic Regression.

I fixed the selection rule before comparing: test F1-score is the main metric, because the classes are imbalanced (62/38) and accuracy alone would reward a model that leans toward predicting "died". Cross-validated F1 checks whether the result is stable, and if the models are within noise, the simpler and more explainable one wins.

- **Higher F1:** 0.724 vs 0.718 on the test set.
- **Higher recall:** 0.667 vs 0.609 — it correctly finds 4 more of the 69 survivors.
- **Stability is a tie:** Random Forest's CV F1 is slightly higher (0.738 vs 0.726), but the gap (0.012) is well under one standard deviation (~0.03), so neither model is reliably better.
- **Explainability:** each coefficient shows direction and strength (e.g. female → higher survival odds; higher class number and older age → lower odds). A 200-tree forest cannot be read this way.
- **Simplicity:** fewer settings to tune, faster to train and predict, less prone to overfitting a small dataset.

What I gave up: Random Forest has higher accuracy (0.816 vs 0.804) and precision (0.875 vs 0.793). If false alarms were the costly error, Random Forest would be the better choice. The decision depends on which mistake matters more — here missing a real survivor was treated as the worse error.

5. Explainability

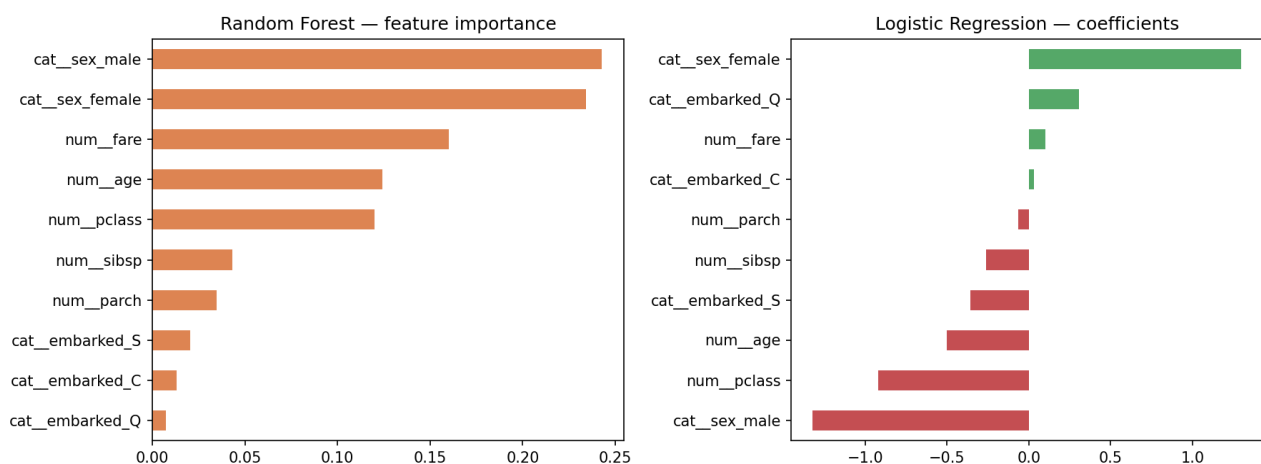


Figure 4. Both models agree on the key drivers: sex first, then passenger class, fare and age. Port of embarkation matters little.

6. Prediction on sample input

`predict_passenger(...)` takes one passenger's details and returns the predicted class plus the survival probability from the selected model. Missing values (e.g. unknown age) are handled by the same pipeline.

Class	Sex	Age	Sib/Sp	Par/Ch	Fare	Port	Prediction	P(survive)
1	female	29	0	0	100.0	C	Survived	0.955
3	male	22	0	0	7.25	S	Died	0.111
2	male	5	1	1	26.0	S	Died	0.353
3	female	missing	1	0	15.5	Q	Survived	0.672

7. Limitations

- **Small test set:** 179 passengers — two or three predictions changing moves the metrics by 1–2 points, which is the size of the gap between the models.
- **Linear model misses interactions:** the 5-year-old boy in the sample gets only 35% survival because the model treats age and sex separately; it cannot fully learn "children first" regardless of sex.
- **Imputation:** 20% of ages are filled with the median, which weakens the age signal.
- **Historical, narrow data:** it reflects 1912 evacuation rules and does not generalise to other situations.
- **Not production-ready:** an 80% score on one split is not enough. There is no external validation, no probability calibration check, no fairness review (the model relies heavily on sex), and no monitoring.

8. Conclusion

Both models perform similarly (~80% accuracy, F1 ~0.72). Logistic Regression is selected because it has the higher F1 and recall on the test set, cross-validation shows no reliable advantage for Random Forest, and its coefficients make every prediction easy to explain. Random Forest remains the better option if precision matters more. Next steps would be tuning both models, adding engineered features (e.g. title from name, family size), and testing on more data before any real use.